

UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Lab #7

Circuit Analysis Lab

PRESENTED BY:

Umer Kashif – 2025-CE-94

PRESENTED TO:

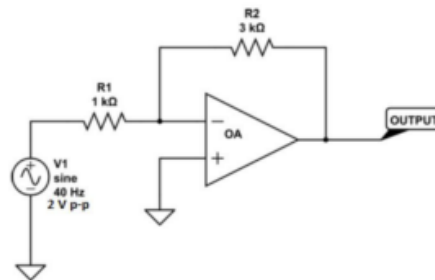
Prof. Mehmood

COMPUTER ENGINEERING – SECTION B

Task:

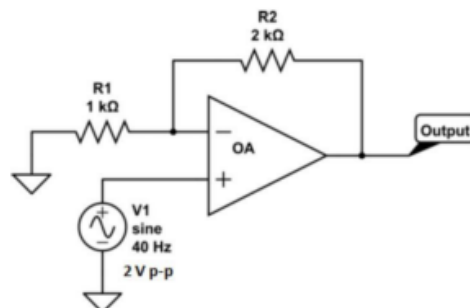
Procedure for Inverting Amplifier:

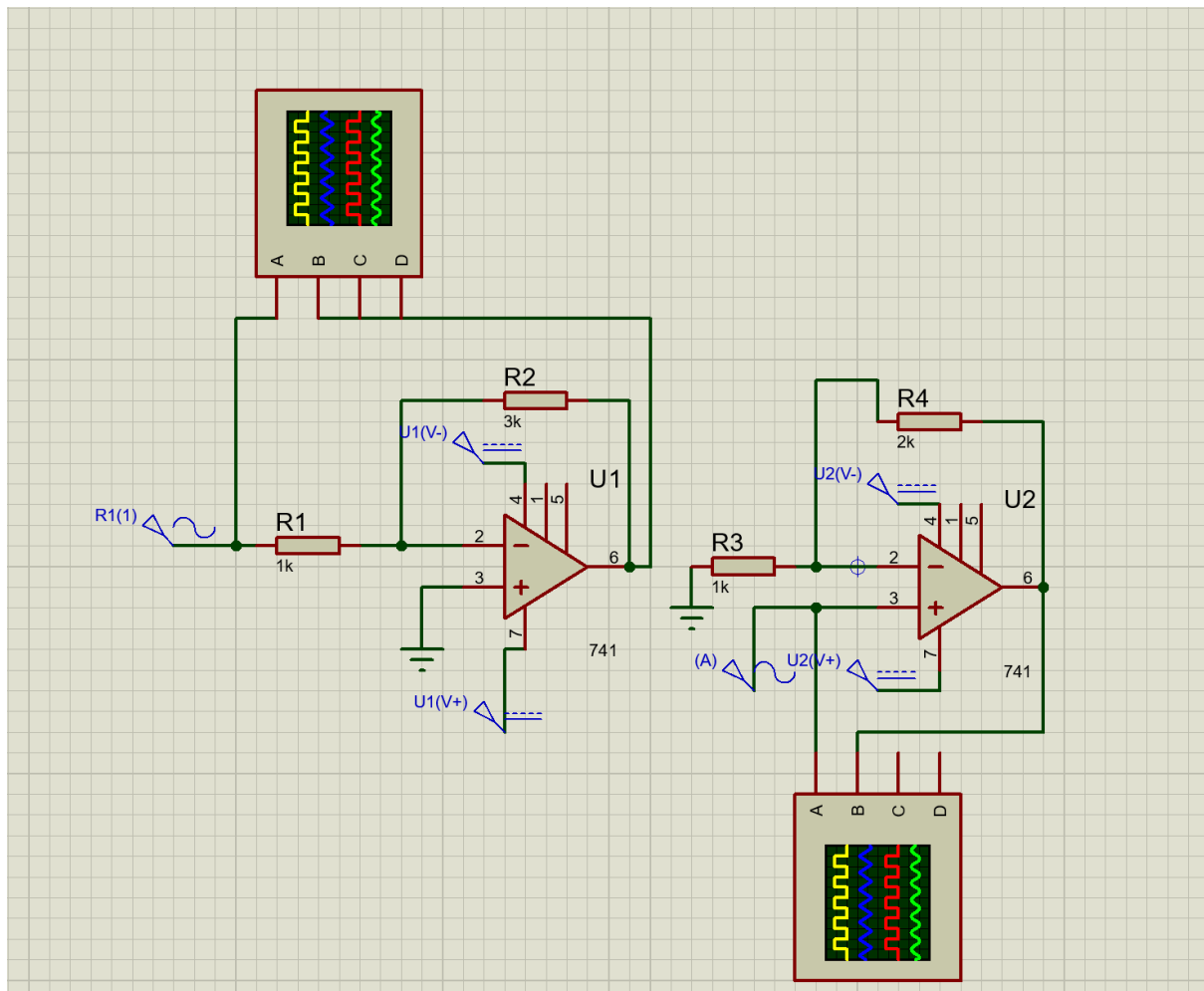
1. Wire up the circuit as shown in the figure below on the breadboard.
2. Give the input to the inverting terminal of the op-amp and simultaneously display the input and output on the oscilloscope.
3. Draw the input & output waveforms at the end of the lab.



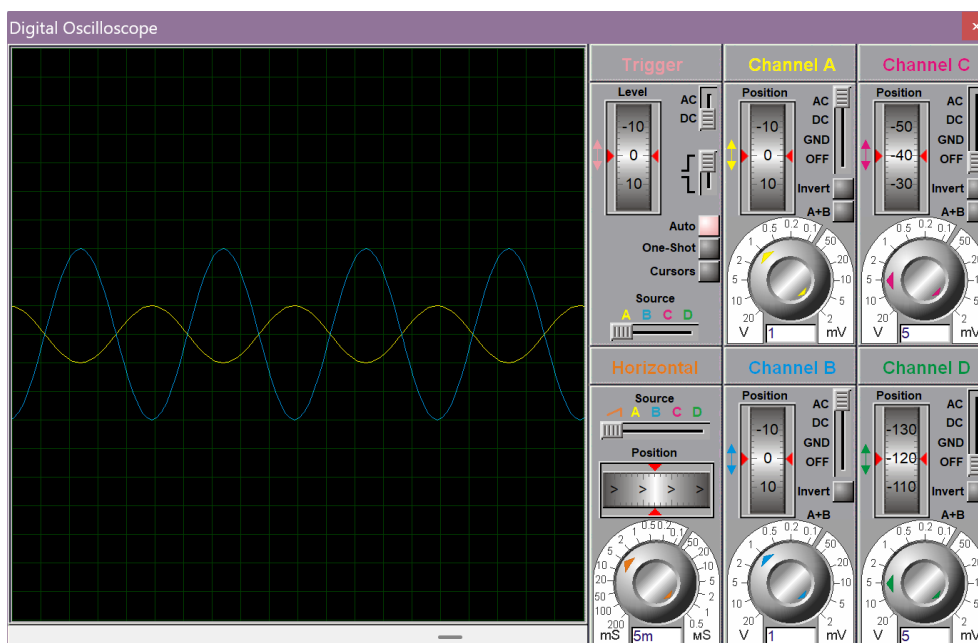
Procedure for Non-Inverting Amplifier:

1. Wire up the circuit as shown in the figure below on the breadboard.
2. Give the input to the non-inverting terminal of the op-amp and simultaneously display the input and output on the oscilloscope.
3. Draw the input & output waveform at the end of the lab.





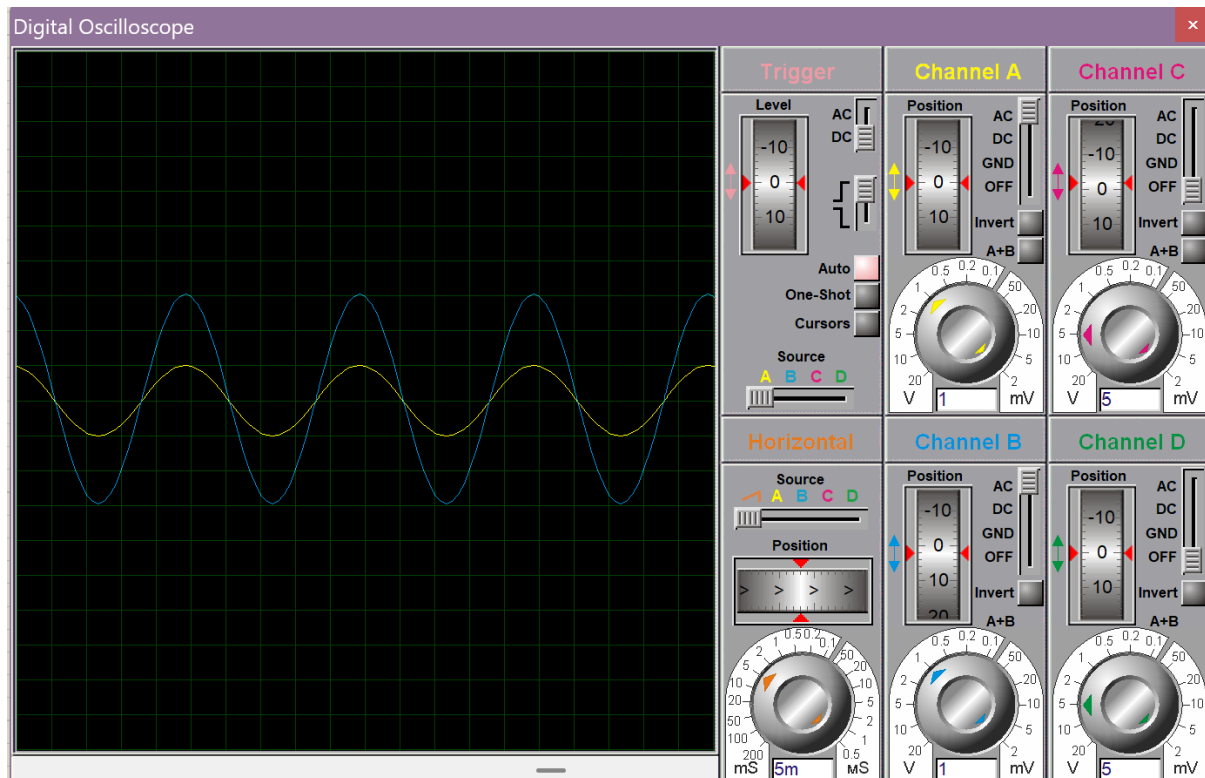
Inverting:



Steps to build in proteus (Inverting Amplifier):

1. In proteus selected the 741 operational amplifier, two resistors, and a ground terminal.
2. Constructed the circuit by connecting $R1 = 1\text{ k}\Omega$ to the inverting terminal (Pin 2) and placing the feedback resistor $R2 = 3\text{ k}\Omega$ between Pin 2 and the output (Pin 6).
3. Grounded the non-inverting terminal (Pin 3).
4. Configured the ac voltage source (Sine generator) to provide a 2V p-p signal by setting the amplitude to 1V with a frequency of 40 Hz, and connected it in series with R1.
5. Powered the op-amp by connecting a +15V DC source to Pin 7 and a -15V DC source to Pin 4.
6. Connected an oscilloscope to observe the input waveform (Channel A) before R1 and the inverted output waveform (Channel B) at Pin 6.

Non Inverting:



Steps to build in proteus (Non-Inverting Amplifier):

1. In proteus selected the 741 operational amplifier, two resistors and a ground terminal.
2. Constructed the circuit by connecting $R3 = 1\text{ k}\Omega$ from the inverting terminal (Pin 2) directly to ground, and placing the feedback resistor $R4 = 2\text{ k}\Omega$ between Pin 2 and the output (Pin 6).
3. Configured the ac voltage source (Sine generator) to provide a 2V p-p signal by setting the amplitude to 1V with a frequency of 40 Hz, and connected it directly to the non-inverting terminal (Pin 3).
4. Powered the op-amp by connecting a +15V DC source to Pin 7 and a -15V DC source to Pin 4.
5. Connected an oscilloscope to observe the input waveform (Channel A) at Pin 3 and the amplified output waveform (Channel B) at Pin 6.

Graphs:

Result:

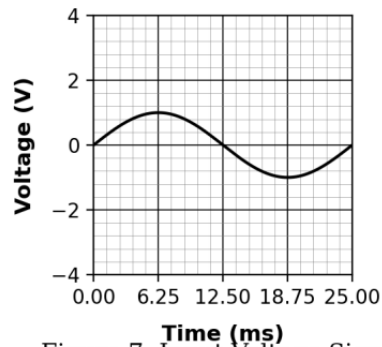


Figure 7: Input Voltage Signal
Scale: X-axis = 6.25 ms/div, Y-axis = 2 V/div

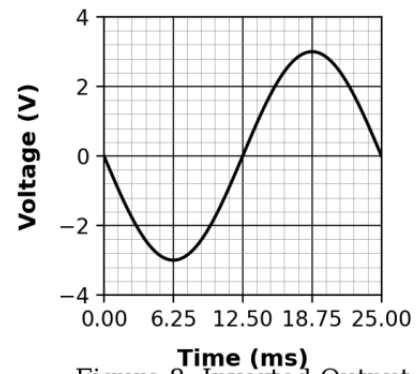


Figure 8: Inverted Output
Scale: X-axis = 6.25 ms/div, Y-axis = 2 V/div

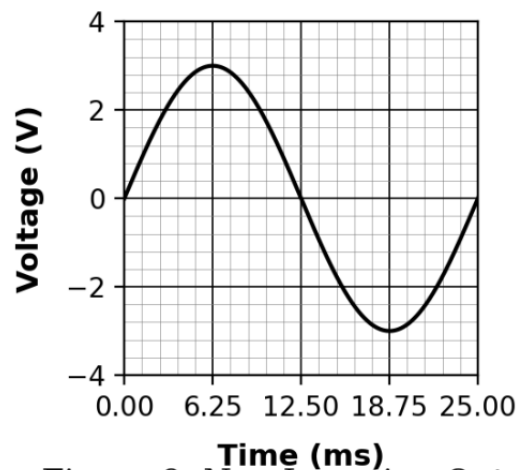


Figure 9: Non-Inverting Output
Scale: X-axis = 6.25 ms/div, Y-axis = 2 V/div